UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

Solve the following equations. Show the complete workflow.

a) 52.738 + AB.2H = --.--------.-.--2

b) 171.2812510 + 1010 1100.112 =
8.

QUESTION 2 (5 marks)

Artificial intelligence (A1) is the computer ability to exhibit human intelligent behaviors such as smart home applications, robots, intelligent tutoring systems and others.

a) In which generation is A1 introduced?

b) What is the programming language used for this technology?

c) What is the advantage of embedding A1 in our everyday life?

d) A1 can also posed as a threat to community. Discuss ONE (1) possible scenario when A1 can harm people.

QUESTION 3 (5 marks)

a) Discuss TWO (2) reasons why the binary numbering system is used instead of a decimal numbering system in computer architecture.

b) Estimate the number of vacuum tubes needed to represent the value of 79 using the binary representation. Show the workflow.

QUESTION 4 (5 marks)

a) Give ONE (1) characteristic of Symmetric Multiprocessor (SMP).

b) Discuss the meaning of "symmetric" in Symmetric Multiprocessor.

c) Give ONE (1) advantage of Symmetric Multiprocessor organization.

QUESTION 5 (10 marks)

a) Why input/output problem occurs?

b) Describe THREE (3) categories for input/output module and provide an appropriate example each.

QUESTION 6 (10 marks)

a) Rank the following memory types based on price ranging from the cheapest to the most expensive.

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Cache, tape back-up, register, hard drive

b) Discuss ONE (1) benefit of cache memory.

c) Differentiate between write through and write back mechanisms in order to store data in cache.

QUESTION 7 (10 marks)

a) Convert -197.007812510 to floating point number in IEEE single precision format in hexadecimal value form.

b) Convert 440AE800h in IEEE single precision floating point number to decimal value form.

QUESTION 8 (10 marks)

a) Label the following device with appropriate input/output module.
Satellite dish
ii.
Thumb drive
iii.
Touch screen
Joy stick
v.
Central processing unit (CPU)
vi.
Modem

b) Define DMA.
Discuss how DMA technique can improve the overall computer performance.

c)

QUESTION 9 (10 marks)

Describe the specificfunction ofdifferent types of registers.

QUESTION 10 (10 marks)

Question 1 is based on the following statement.
In 1942, Mr. X and Mr. Y proposed an all-electronic calculating machine. The U.S. Army, meanwhile, needed to calculate complex wartime ballistics tables. The result was Electronic Numerical Integrator And Computer (ENIAC) that was built between 1943 and 1945. ENIAC is the first large-scale computer to run at electronic speed without being slowed by any mechanical parts. For a decade, until a 1955 lightning strike, ENIAC may have run more calculations than all mankind had done up to that point.

a) In which computer generation, ENIAC is developed?

b) Who built the ENIAC machine?

c) What is the main technology used in the ENIAC machine?

d) Outline a possible solution for each of the following problems that are faced when ENIAC is used.

Problem	Solution
Produced a lot of heat	
Consume a lot of ener	
Lar e in size	
Slow rocessin	
Limited ro rammin ca abilities	

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PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 1 and Table 1.
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.

$$z = A^2 - B * C + \frac{2}{D + 5}$$

Figure 1

Table 1: Machine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	suB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	